MANASWINI ZAGABATHUNI

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EDUCATION

Indiana University Bloomington — United States

Master's in Computer Science, **GPA: 3.78/4.0** — 2025-2027

Relevant Coursework: Applied Machine Learning, Applied Algorithms, Software Engineering, Applied Database Technologies, Large Language Models, Systems and Protocol Security and Information Assurance, Computer Networks, Database Design, Security for Networked Systems.

SRM University — Chennai, India

Bachelor of Computer Science and Engineering, **GPA: 8.96/10** — 2021-2025

(Team Lead for Minor and Major Projects)

Relevant Coursework: Problem Solving, Data Structures and Algorithms, *Object-Oriented Design and Programming, Design and Analysis of Algorithms, Operating Systems, Data Mining and Analytics, Compiler Design, Artificial Intelligence.*

WORK EXPERIENCE

Teaching Assistant - Biomedical Data Science in Practice — May 2026 - Aug 2026

- Assisted graduate students in applying Python, data science, and machine learning techniques to biomedical research using the NIH All of Us Research Program, including clinical, socioeconomic, wearable, and EHR data.
- Guided students through quizzes, homework, and final research projects, providing technical support in data wrangling, visualization, statistical analysis, and machine learning using the All of Us Research Workbench.
- Conducted office hours and troubleshot data access and analysis issues while promoting ethical data handling.

AICTE - *AWS AI-ML Virtual Intern* — Jan 2024 - Mar 2024

- Deployed machine learning models on AWS EC2 and S3, improving deployment efficiency by 30%.
- Architected scalable cloud-based ML pipelines enabling automated training and real-time inference.
- Automated cloud environment configuration, reducing setup time and manual errors.

SKILLS AND INTERESTS

Technical Skills: *Python, C, C++, JavaScript, SQL, Dart, TensorFlow, LLMs, RAG, Vision Transformers, EfficientNet, React, Flask, FastAPI, REST APIs, MySQL, MongoDB, Supabase, ChromaDB, Flutter, Firebase, AWS EC2, AWS S3, Docker, Linux, Git, GitHub, Manim*

Certifications: SQL Basic (Hacker rank), Database Foundations (Oracle Academy), Building Video AI Applications at the Edge on Jetson Nano (NVIDIA), Programming for Everybody(Coursera), MongoDB CRUD Operations in Python (MongoDB).

ACADEMIC PROJECTS

GeneGardenAI - Evidence-Grounded Biomedical Discovery Platform — Oct 2026

- Built a full-stack biomedical discovery platform integrating gene–disease associations from Open Targets and scientific literature from Europe PMC into a provenance-aware PostgreSQL knowledge graph.
- Developed a FastAPI backend and React/TypeScript interface for graph-based discovery, candidate-path ranking, and evidence exploration, with automated backend, frontend, and end-to-end testing.

InsightFlow: Full-Stack Data Analytics & Machine Learning Platform — Sep 2026

- Developed a full-stack analytics platform for dataset profiling, data quality assessment, interactive visualization, and machine learning workflows.
- Implemented classification, regression, and clustering workflows with model evaluation and SHAP-based explainability to help interpret model predictions.
- Strengthened application security with HttpOnly cookies, CSRF protection, API rate limiting, and automated testing across backend and frontend components.

VisualAIze - LLM Pipeline for Animated Math Explainer Videos — June 2026

- Built a multi-agent LLM pipeline that transforms math topics or PDFs into narrated educational videos, using six specialized agents for document parsing, concept extraction, lesson design, scene planning, and Manim code generation.
- Developed a ChromaDB RAG knowledge base with ~9,000 math documents to improve factual grounding and reduce hallucinations.
- Deployed the backend using FastAPI and Docker, enabling an end-to-end pipeline for automated educational video generation.

ACHIEVEMENTS

Published Research paper in IJCRT, National Service Scheme (NSS) (Volunteer Experience - 01/2022 to 06/2022), Participated in Building Video AI Applications at the Edge on Jetson Nano workshop by NVIDIA.

PUBLICATION

- Tomato Leaves Disease Classification Using Vision Transformers and EfficientNet- *https://www.atlantis-press.com/proceedings/icisd-25/126017029*
- *A Cross-Platform Flutter Chat Application Using Firebase and Supabase - https://www.ijcrt.org/papers/IJCRT2410189.pdf*